

UNIVERSITY OF
EASTERN FINLAND

Enhancing user experience in AR applications: A design approach for user interface

Sachin Basnet
IMPIT
University of Eastern Finland
Faculty of Science and Forestry
School of Computing Computer Science
2024

University of Eastern Finland, Faculty of Science and Forestry, Joensuu School of Computing
Department: Computer Science
Degree Programme: International Masters Programme in Information Technology
Basnet Sachin: Enhancing user experience in AR applications: A design approach for user interface
Master's Thesis, 45 pages.
Thesis Supervisor: Markku Tukiainen
December 2024

Abstract

Augmented Reality (AR) is an emerging technology which has been setting up its powerful strength in various fields. However, many AR applications suffer from bad user experience because of ineffectual user interface (UI) designs. The main objective of this research paper is to explore the existing design practices, point out their challenges and propose forward thinking design plans for AR UIs to improve usability, functionality and overall user satisfaction. Through user interview and online surveys among potential users and designers, this study identifies and understand their problems while crafting and using such applications. Finally, the result suggests that user interface for AR apps should be designed by adopting user's needs and preferences which could potentially decrease the cognitive load and increase the engagement of users with the system. The findings render obtainable recommendations for designers, developers and researchers for future creation of UIs.

Keywords: augmented reality, user interface design, user experience

Foreword

This thesis was done at the end of my master of IT degree at the School of Computing, University of Eastern Finland.

Many challenges appeared during the process of writing this thesis and delivering the project. Completing this thesis was significantly harder than I thought. I thank everyone connected to my journey for their significant roles in several stages. I thank my supervisor Prof. Markku Tukiainen for the support he gave and his patience as well as the freedom he allowed me to develop this project as I imagined it.

I would like to thank all my friends who connected me and helped me go through the interview process. I would like to express my gratitude to my supervisors who believed in me and encouraged me to select this topic. Lastly, I have to thank my family for supporting me during all my lifetime with only amazing patience.

Abbreviations

AR	Augment Reality
UI	User interface
UX	User experience
UCD	User-centred design

List of Figures

Figure 1: Age distribution of survey respondents 27
Figure 2: AR application usage frequency 28
Figure 3: Most frequently used AR applications 29

List of Tables

Table 1: Age distribution and Usage frequency of AR application 28

Table 2: User preferences for AR interface elements..... 30

Table 3: Average importance ratings of AR features 30

Contents

1 Introduction 7

1.1 Problem definition 7
1.2 Research objective and questions 8
1.3 Research methodology 9
1.4 Structure of the thesis 9

2 Background 11

2.1 Overview of augmented reality technology 11
2.2 UX design principles in AR applications 12
2.3 Challenges in AR user interface design 13
2.4 Design frameworks and approaches for AR UI 15

3 Research Method 18

3.1 User-centered Design Approach 18
3.2 Research Context 18
3.3 Data Collection 19

3.3.1 Interview 19
3.3.2 Online Surveys 20

3.4 Data Analysis 20
3.5 Validity and Reliability 21
3.6 Ethical Considerations 21

4 Findings 22

4.1 Qualitative findings from Interviews 22

4.1.1 Theme category 1: Interface usability 22

4.1.2 Theme category 2: Design principles in AR UIs 22
4.1.3 Theme category 3: Challenges faced by users and designers 23
4.1.4 Theme category 4: Feedback and design solutions 25

4.2 Quantitative findings from User Surveys 26

4.2.1 Demographic info 26
4.2.2 Most frequently used AR applications and their usage patterns 27
4.2.3 AR Interface Elements 29
4.2.4 Feature Importance 30
4.2.5 User Challenges 31

5 Discussion 33

5.1 Current user interface design principles in AR applications 33
5.2 Challenges face by users with existing AR UIs 34
5.3 Design approaches or solutions 36

6 Conclusions 38

6.1 Research Contribution 38
6.2 Limitation 39
6.3 Future Work 39

7 References 41
8 Appendices 44

1 Introduction

Augmented Reality (AR) is a technology that enables us to overlay virtual content onto the real world (R. Silva, n.d.). Basically, it creates some kind of illusion to human eyes where the content is actually present in the physical environment. AR essentially allows us to merge the digital and physical worlds. It was introduced in 1992 by Thomas Preston Caudell, who build applications to view industrial diagrams (Arena et al., 2022). By then, AR technology has rapidly advanced where we can find applications in various fields such as education, healthcare, retail, and entertainment. As the AR applications continue to grow, it is very important to prioritize the user experience to ensure seamless and engaging interactions. But designing user interfaces (UIs) for AR can be tricky. The traditional rules for UI design do not fit well with AR. Such as that, designers have some challenges to face. The area of making user-friendly designs for AR is still quite new and not fully explored yet. This is due to the vastness and complexity of UI design in AR. Designers often lack clear design guidelines or system to help them build great AR experiences. This presents unique challenges, such as maintaining spatial awareness, minimizing cognitive load, and ensuring intuitive user interactions. This research paper explores design principles and practices for user interfaces in AR applications that can enhance user experiences, identifying areas for improvement and proposing new strategies for creating effective AR UIs.

1.1 Problem definition

Augmented Reality applications are becoming more common and available at present. This technology is gradually transforming the way we live, work and play. Many big tech companies such as Apple, Google, and Microsoft have been working from early 2000's for building different AR devices that could facilitate the users to interact with digital content over physical surroundings. As of now, it has widely applied on various domains including gaming, healthcare, education and many more. However, as AR continues to evolve, there is extreme need for well-designed user interfaces to ensure seamless and intuitive interaction between users and AR applications. Despite the fastest adoption and potential of AR technologies, many users and developers face on major challenges, that is the design of intuitive and effective user interfaces. AR applications need users to interact with virtual objects in a real-world environment, yet many

existing UIs struggle to provide intuitive navigation which could potentially lead to user frustration and cognitive overload. Users often find it difficult to manage tasks, understand virtual cues, or interact with content in a natural and fluid manner. Poorly designed AR UIs not only hinder user experience but also limit the widespread adoption of AR technologies. Users may experience discomfort, confusion, and inefficiency, which diminishes the overall potential of AR applications. Additionally, designers face challenges in creating UIs that align with the dynamic, spatial nature of AR environments, as traditional 2D design principles do not fully address these complexities.

Although there have been advancements in UX design principles for traditional applications, there is still a significant gap in understanding how to apply these principles effectively in AR environments. Many AR applications continue to rely on traditional UI elements that may not translate well into immersive, spatial experiences. As a result, there is an urgent need to develop tailored design approaches that enhance user experience in AR.

1.2 Research objective and questions

This thesis aims to address these challenges by investigating current AR UI design practices, identifying areas for improvement, and proposing new design approaches that enhance the overall user experience.

Considering the goals and problem definition, we aim to explore the following research questions:

RQ1: What are the current design principles applied in AR user interfaces?
RQ2: What are the common challenges and pain points users face with existing AR UIs?
RQ3: What new design approaches or solutions can be proposed for AR UIs?

1.3 Research methodology

For this study, a mixed method has been used. A qualitative interview was conducted among designers, industry experts. Similarly, online survey was carried out targeting users of AR applications. The interviews will give insights about the challenges of AR user interface design, while the surveys will assess user satisfaction and their needs regarding AR applications. The sampling strategy focuses on 5-10 professionals for interview and approximately 50 AR users which will be completing survey. The interview questions will be semi-structured which will mostly focused design challenges, current UI design approaches, and user feedback loops. The online surveys will use combinations of multiple choice and an open-ended question to collect user feedback and to understand their pain points. Interview sessions are planned for August 2024 and survey collection will follow shortly after.

Before conducting the interviews, I have participated in an AR/VR design course offered by ELVTR to gain a deep understanding of AR technology and user interface design. The interview was recorded, transcribed, and translated into English where necessary. For survey, google form was used as a platform to gather data and ease of use. After collecting data, they were analysed using descriptive statistics and cross-tabulation to understand users satisfaction levels and the relationships between demographic factors and AR usage. By the combination of user feedback, designers and industry expert's insights, and innovative design solutions, this research paper is looking forward to contributing the creation of more effective and intuitive AR interfaces.

1.4 Structure of the thesis

Chapter 2: Background provides the theoretical foundation of the thesis. This article reviews similar literature on AR technology, UX design principles in AR, challenges and best practices within the AR UIs design.

Chapter 3: Research methodology outlines the research approach, research context, and data collection and data analysis method.

Chapter 4: Findings analyses the data acquired for this research.

Chapter 5: Discussion goes back to the research questions and answers them with the findings from chapter 4.

Chapter 6: Conclusion summarizes the research findings and suggests areas for future work.

2 Background

2.1 Overview of augmented reality technology

AR technology places computer generated images, animations and other digital assets on the top of the physical world. It is one of the most significant achievements in today's digital world, that has been greatly improving the user experience in different fields, especially in education, culture and industry. In the educational sector, AR has been seen to improve the level of learning experience by making it more engaging for the students. For example, research shows that AR enhances traditional way of teaching by overlaying great visual representation of concepts in real time making learning more efficient and engaging (Nigam & C, 2022; Wen et al., 2023). Likewise, AR applications has been widely used in subjects like history and architecture, where students can interact with three dimensional models of historical scenes (Challenor & Ma, 2019; Putro & Setyowati, 2022). This approach of learning helps the student to grasp the knowledge while also creates interest and motivates them for exploring new things within the subject matter (Challenor & Ma, 2019). Furthermore, the advancement of AR has resulted in enhanced software applications which incorporates the use of mobile devices, smart glasses to interact with augmented content without using their hands. The use of AR in cultural heritage sites has helped in tourism sectors by allowing visitors to engaged with history (Bracho Mosquera, 2024; Jung & tom Dieck, 2017). Thus, the evolution of AR technology has offered improved augmented experiences from simple to complex system (Billinghurst et al., 2015). During this timeline there has been a major improvement in computer graphics, sensors, mobile computing and many more. So that AR applications are becoming more applicable in various fields (Alkhamisi & Monowar, 2013). The AR technology is still in its growing phase. The future research work will improve the user interface and address some of the existing issues of AR user interfaces and content development (Lim & Lim, 2020).

2.2 UX design principles in AR applications

The principles of user experience (UX) design in augmented reality applications are crucial for crafting engaging, intuitive, and effective user interfaces. Since the AR technology is evolving, UX field has become an essential area to focus on and it is equally important to understand design practices for developers and designers. According to present research, one of the most important guidelines of UX design in AR applications is to create as simple interfaces as possible that enable easy interaction between the user and the content. Cruz-Carrizales et al. emphasise that in AR, the user interface description languages should consider the characteristics of AR environment to enable proper user interaction and understanding (Cruz-Carrizales et al., 2021). Additionally, Öztürk explains that the AR interfaces can enhance artificial intelligence systems if there is a good interface design that can have a positive impact on usability and user satisfaction by giving feedback to the users when they are interacting with artificial systems (Öztürk & Genç, 2023). Similarly, these findings also align with the thoughts presented by Hammady et al., who has developed a UX design model for cultural heritage that emphasizes the improvement of user interaction in AR applications. This study has shown that proper design will improve interaction in the overall AR applications.

The other important aspect of UX design in AR is the use of context aware interface that can capture information from the surrounding environment and the actions of the users. Grubert et al, argue that context-awareness plays a crucial role in AR. As it makes it possible to provide the user with exactly the information that is useful or interesting in the given situation and time (Grubert et al., 2017). This is further supported by Li and Cheng where they stated that AR technology can make communication and information sharing more vivid and rich due to the real time and context-based information that is delivered to the users (Li & Cheng, 2023). Adaptive interfaces are not only good for user engagement, but also help the user to gain better insight about the content presented. For example, the author Tenh and Shiratuddin argued that adaptive AR models can provide better user experience through adaptive content presentations (Tenh & Shiratuddin, 2022).

Furthermore, in the case of AR applications, learnability and memorability must be taken into consideration while designing a particular application in order to support easy navigation and usage. Ejaz et al. highlights several design principles which are very suitable and applicable for creating interfaces that users can easily learn and recall. Such as, simplicity, feedback, and visibility (Ejaz et al., 2019). This is particularly important in AR applications, where users might be interacting with complex digital augmentations in the real environment. The research by Davidavičienė et al. further supports this notion, where they emphasize that to evaluate user experience in the AR applications, it is necessary to apply the evaluation methods. Additionally, they stress that more attention should be paid to the user-centered design approach as it benefits the usability and satisfaction levels (Davidavičienė et al., 2020).

In addition to these principles, the integration of interactive design elements is vital for enhancing user experience in AR applications (Md Nor & Abdul Razak, 2021). This interactive process is supported by the research of, who suggest that AR can dramatically improve the consumer decision-making process by creating immersive and interacting experiences for exploring products (Pantano et al., 2017). These kinds of interactive features are not only more fun, but they also create a deeper connection between the user and the content. Moreover, feedback mechanisms in AR apps are crucial. Good feedback helps people learn what they are doing in response to their actions and walk properly through the interaction process. Further it was also supported by the research which explains that good UI design must include methods to provide feedback in order to enhance the learning experience in AR environments (Williams et al., 2020).

2.3 Challenges in AR user interface design

The design of user interfaces for AR applications consists of set of unique challenges that spans from blending virtual and physical world to achieving seamless user experience. This section outlines the current challenges faced in AR UI design from various recent studies.

One of the primary challenges in AR UI design is the integration of virtual elements into real-world environments. It requires an effective interface design to prevent users from being overwhelmed or confused by AR features. Kiourexidou et al. highlights the importance of user experience in educational AR apps. They stated that the interface should not only be intuitive, but it has to be engaging and easy to navigate (Kiourexidou et al., 2024). Similarly, Ejaz et al. address the importance of interfaces that are not only user-friendly but should be self-explanatory (Ejaz et al., 2019). This issue was further discussed by Lages and Bowman, who suggested that the UIs for AR system must accommodate the screen or dynamic features based on physical settings and user's preferences (Lages & Bowman, 2019).

Another key problem is the cognitive load that user face while interacting with AR interfaces. Teng et al. presents machine learning model to predict cognitive load in AR head-up displays (HUDs). They have suggested the innovative methods for optimizing the interface design which could possibly reduce the cognitive load (Teng et al., 2023). According to Öztürk, AR can enhance the usability of AI systems by offering intuitive interfaces for complex decision-making processes (Öztürk & Genç, 2023). This is especially important in applications that need spatial data processing. From here it has become clearer that the challenges lie in designing interfaces that present information clearly without overwhelming the user, which is vital for maintaining engagement and ensuring effective learning outcomes.

Usability testing is another critical aspect of AR UI design that presents challenges. Several studies have found that the traditional method of usability testing may not fully captured the unique interactions in AR applications. For example, the system usability scale (SUS) and handheld augmented reality usability scale (HARUS) are useful tools for evaluating usability. However, their usefulness may vary depending on the context of AR apps (Vidak et al., 2024). Furthermore, combining multimodal interactions such as gesture and voice commands may complicate usability evaluation since these interactions must be intuitive and responsive to user needs (Giariskanis et al., 2022).

Moreover, people often complain about the discomfort they have experienced from AR devices such as headsets and glasses. This is another serious problem that could negatively impact the user experience. According to Alzahrani, bulky and expensive augmented reality gadgets can cause discomfort, which can indirectly hamper in user engagement and learning (Alzahrani, 2020). This was even supported by Altinpulluk et al., who has also emphasized that there should be universal design principles for designing and manufacturing such products that could increase usability and reduce discomfort among all users (Altınpulluk et al., 2019). At last, creating user interfaces for augmented reality applications needs multidisciplinary approach that combines the insights from user experience design, cognitive psychology, and technology development.

2.4 Design frameworks and approaches for AR UI

As of now, there is a no such standard design guidelines available in the augmented reality user interface design field (Blokša, 2017). Currently, the most utilized user interfaces and design principles are mostly focused on specific devices, such as Google glass, Microsoft HoloLens and many more. Thus, we can easily find developer guide that are primarily focused on computer games and virtual reality (VR) games (Blokša, 2017). There are five core design elements proposed by Blokša which is particularly useful for designing the UIs of AR application (Blokša, 2017). The first element is Presentation. While designing and sharing the content, one should consider the ratio of content to the environmental space. This means, all the content should be displayed and positioned with respect to natural viewing area. On top of this, Bloksa suggested to use three dimensional buttons for all applications. In addition, he also said that responsive design can increase the usability and visibility of the applications. The second element is Control. The control should be more of different types with an option of adjusting based on the environmental scenario. Third element is displaying instructions. It is suggested to offer multiple instructions to the users with the aim of educating different level of users and allowing them to skip or close those displayed instructions independently. Fourth element is Navigation. When designing user interface, designer usually give emphasis to layout, colors and fonts. It is noted that the same amount of emphasis should also be placed on navigation. Navigation should be

intuitive and easy to follow, and in general the interface should not be confusing. The last feature is Notifications. Notifications are used to make the user up to date about what's happening or what will happen within an application. With that said, user should be given full control to customize the way they are notified and should be able to select the type of notification that they want. However, these set of guidelines are concerned with general principles only. They do not cover interaction and control types such as voice, gesture and multimodal controls.

Furthermore, Shukri et al. suggested a user interface design guidelines for designing a mobile AR travel app. These guidelines were introduced by incorporating PAD theory of user's emotion, i.e. pleasure, arousal, and dominance. This design approach was particularly based on four main aspects. They are, 1) applying context to deliver content 2) minimizing cognitive load 3) to enable easy interaction and comfort navigation for user 4) to allow 3D interaction. However, these approaches do not have appropriate process for assessing the usability of corresponding applications.

Likewise, based on cognitive load theory (CLT), Oluwadunsin Dabor et al. have developed a user interface guideline. They have integrated the design principles proposed by Shneiderman for desktop applications, human computer interaction, and the rules for designing AR systems and a theoretical model for managing psychological workload within the work environment. They have shaded lights on mainly four different areas. Such as, 1) UI should be suitable for global accessibility and must meet user preferences 2) allow users to control the content 3) reduce the memory load 4) context should be used for sharing information. However, this guideline lacks the proper evaluation validation is not available yet.

Moreover, in Augmented Reality (AR), the users may experience perceptual difficulties as they try to perceive and interpret data. Most of these problems are about alternation in the information, adaptation of abstraction, object ordering and visibility. In order to solve these problems, Cledja Rolim et al. suggested visual design guidelines for AR instructions that aim at AR visualization. Those methods are, 1) Directional cueing: that instructions should point in the direction of

correct movement and in some cases with respect to velocity or acceleration. 2) Emphasize the part to be moved or changed: this is especially true in the case of body instructions which unambiguously tell the user which part of the body should be transferred. 3) Allow for a diversity of visual appearance attributes: AR instructions should be visually presented based on their surroundings. Users should have control over the display of commands, as some jobs may need precise patterns to follow. 4) Provide real-time feedback methods must be included into the command process so that the user understands if he or she is doing something correctly, whether corrections are needed, or whether alternatives exist. This input should be supplied in real time. 5) Managing Occlusion and depth relationships: when transmitting 3D data, such as disassembly tasks, Instructions must contain visual clues to assist the user comprehend the occlusion connections and depth. The above design guidelines are meant to enhance the readability and effectiveness of AR instructions, which leads to improvement in the general user experience.

3 Research Method

After thorough study of literature and keeping them in mind, this paper aims to explore how the users interact with the user interfaces in the AR applications and how can we enhance the overall user experience. Specifically, it is very important to understand the challenges in AR interfaces and how they affect usability, interaction, and user satisfaction. In this chapter, we discuss the research context, data collection methods, and analysis techniques.

3.1 User-centered Design Approach

This research follows a user-centered design (UCD) approach with a strong focus on qualitative methods. According to (Nielsen J, 1993), user-centered design seeks to involve end-users in the design process to ensure that the interface meets their needs and enhances usability. This research adopts this approach to understand user's experiences with existing AR interfaces and gather feedback on ways to improve those interfaces.

Following this method, we will explore, 1) how users interact with AR interfaces 2) what challenges users face with the current AR UI designs 3) how we can enhance user experience in AR application.

3.2 Research Context

AR technology is advancing rapidly by creating new opportunities for immersive experiences with products like the Apple Vision Pro. However, designing effective user interface for AR is still tough. This study focuses on understanding these challenges and aims to improve the design of AR UIs by gathering insights from users and designers working in this space.
Following questions are aimed to explored:
RQ1: What are the current design principles applied in AR user interfaces?
RQ2: What are the common challenges and pain points users face with existing AR UIs?
RQ3: What new design approaches or solutions can be proposed for AR UIs?

To explore these questions, we selected participants with direct experience using AR applications, either as end-users or as UI designers involved in creating AR interfaces. Since, I have participated in AR/VR design course virtually which was offered by ELVTR. Throught this course, I have had privilege of connecting with likeminded individuals and AR design enthusiasts. Additionally, I attended events organized by Helsinki XR Center, which was held in August, where I got chance to meet industry experts. This experience not only helped me to gain knowledge of AR design but also provided access to valuable contacts for my research. From there, I opened up with them and shared things about my research and some of them were very happy by seeing my eagerness to learn and contribute to AR design space. I asked them whether we could possibly setup for short interview and share their thoughts and they were ready to be a part of research. Similarly, for survey, it was quite easy for reaching out the target audience through social media platforms, such as WhatsApp group. I distributed the survey link across multiple student group and received good number of responses. These methods allowed me to engage with well-rounded group of individuals and achieved diverse results.

3.3 Data Collection

3.3.1 Interview

The semi-structured interviews were conducted with 5 AR designers and experts during August 2024. Each interview was conducted online and as well as in person. The interview time was set to 45 minutes. The interview was recorded and later, it was transcribed.

Semi-structured interviews are well popular type of interview. It allows the interviewer to guide the discussion while giving participants the freedom to elaborate on their experiences (Galletta, 2013).The interviews explored user challenges with AR interfaces, the usability of various UI components, and potential improvements. The questions pattern was open-ended designed to encourage in-depth responses and to directly address the study's research questions.

3.3.2 Online Surveys

In addition to interviews, an online survey was conducted among AR application users using Google Forms. Initially, the survey starts with demographic questions such as age, occupation, experience with AR. It is then followed by a mixed of Likert scale questions, multiple choice and open-ended questions to understand user's satisfaction with current AR interfaces. In this survey, the age group of targeted audience was between 18 to 50 years, who are familiar with AR applications on mobile and headset devices. This process helps us to reach a larger audience and gather quantitative data on user experience.

3.4 Data Analysis

The transcript of recorded interviews was analysed using thematic coding. Thematic analysis is a process of finding common or repeated theme from the data which is very suitable for qualitative research (Robson, 2011). Basically, thematic methods involved the following steps described by Robson:

- Familiarization: Initially the recorded interviews were transcribed and read properly to identify main motive.
- Coding: Secondly, after finding common theme or idea from recordings we categorized them based on similarity.
- Theme identification: Once we organize in a group we identify theme. For instance, we have organized the codes for AR UI challenges such as spatial interaction, feedback mechanisms, and visual comfort.
- Refinement: We revisited themes to make sure they are presented accurately.

Similarly, we have analysed the survey data using descriptive statistics method. Our primary goal was to understand user's pain points, their preferences and satisfaction level. Once we have the data, we find out key metrics, such as ease of use, clarity of design, and visual impact.

3.5 Validity and Reliability

The validity and reliability of qualitative data is difficult to establish. However, three different measures were implemented which was devised by Robson (Robson, 2011).

- Triangulation: The answers from interviewee and the response from survey were compared to check the consistency in the challenges and design solutions provided by nerd users and experts.
- Member checks: After transcribing the interviews, participants were requested to review their transcripts to ensure they are presented accurately.
- Diversity in participants: Selecting participant is very critical. This research includes diverse range of participants such as both novice and expert end-users, AR/UX designers, AR experts. This has helped us to understand broad perspectives which has increased the generalizability of our findings.

3.6 Ethical Considerations

Before conducting interviews and online survey all the participants were informed about the research goal clearly. They were free to answer the questions based on their understanding and own experience. Additionally, consent form was provided which clearly outlines the purpose of this research, told them that participation was voluntary. The data collected were stored securely and all identifying information was anonymized during transcription and analysis.

4 Findings

The data for this research was collected from two sources. They are, qualitative interviews with AR/UX designers and quantitative surveys completed by AR application users. This chapter presents the findings from both data sources.

4.1 Qualitative findings from Interviews

4.1.1 Theme category 1: Interface usability

Usability is an important aspect in UI design space and the same applies to AR applications. There have been several advancements in designing and developing digital products, but one key point always remains same, i.e. how easily users complete certain task within a website or app. Likewise, when we conduct the interviews, majority of designers, developers and professionals frequently mentioned the issues of interface usability. They have mentioned that AR interfaces tend to become cluttered, which makes difficult for users to navigate and complete tasks efficiently.

"Most AR interfaces are over-crowded with visual elements. Novice users get lost or confused because of poor navigation and especially when users have to deal with too many UI elements overlap with real-world objects." (S3)

"We often hear complaints that the AR interfaces are too complex because in some applications, there's too much information is displayed at once and as a result it frustrated users from making quick decision." (S1)

4.1.2 Theme category 2: Design principles in AR UIs

While having discussion on current design practices, most participants had underlined the importance of minimalism, spatial awareness and real-time feedback in AR design. They also

suggest that these are the core aspects that one should consider while designing user interfaces for AR applications. This will definitely help to create seamless and smooth user experiences in AR environments. After this one thing has been cleared, i.e. Simplicity the best. Most AR designers highlighted the need for simplicity, where only the essential content is displayed on the screen.

"Too many options or visual elements is a big no. Giving too many options can create confusion and delay in making decision. Thus, we aim to keep things minimal and purposeful." (S2)

Another aspect of UI design is spatial awareness. One interviewee suggested that good spatial design allows people to feel connected to the virtual and physical world.

"It is very important to understand the user's feeling and emotion. With that in mind, we try in many ways to design all the graphics elements to feel as realistic as if they belong in the physical world." (S3)
"Designers often take inspiration from traditional UI design system, but it is very crucial to educate and adapt themselves with 3D graphics design and spatial positioning." (S1)

The last but not the least, real-time feedback is an important principle of UI design in AR system. Mostly, people use different devices to experience the immersive technology and perform certain tasks. Many users get lost and confused while exploring app's features or screen in AR application. Therefore, quick response to user actions are equally important to ensure smooth experience.

"Immediate feedback helps users to navigate the interface effectively." (S4)

4.1.3 Theme category 3: Challenges faced by users and designers

Both users and designers face numerous challenges when interacting with AR UIs. In interviews, we focus more on understanding AR expert's perspective; what they actually think the typical

problems users face, and the challenges designers have to go through while developing AR UI. The following challenges were consistently highlighted and more common in interviews:

Gesture recognition issues: Many AR applications depend on gesture-based interactions, but participants mentioned that gesture controls are unreliable. Sometime users may find it difficult to interact the interface or they have to repeat actions due to inconsistent and unresponsive gestures. This could be more problematic when user tries to interact in unsuitable environments such as crowded place, bright sunlight, motion related activities like walking, running etc. In such situations, gestures may not be recognized properly.

"It has been difficult to make gesture controls sufficiently responsive for all users. However, recently apple vision pro has done brilliant job. They have implemented advance machine learning algorithms for tracking hands and eyes precisely and made significant improvement in integrating hardware equipment, offering seamless experience in complex environment." (S5)

Users experience variability: Another significant challenge in AR UI design is wide range of user base with different levels of knowledge. It is obvious that different users come with different level of familiarity and comfort with technology. This could lead to inconsistencies in how they engage with AR applications. This has become challenging to create one screen that could fit for all users.

"A seasoned gamer might find an AR game intuitive and engaging, while a less experienced user may struggle to understand the features and navigation process." (S4)

Cognitive load and multitasking: While using AR applications, user often have to deal with both virtual object and physical objects. This dual focus at a time can be challenging to process information accurately and effectively.

"We need to be very careful about how much information we display at once. Also, we should provide a customization feature so that user can have full control over what they want to see and how many tasks they can handle." (S5)

4.1.4 Theme category 4: Feedback and design solutions

Participants also presented their point of view regarding user-centric design solutions to improve the user experience in AR application. Here are the main common areas that interviewee focused on:

Enhancing user onboarding and tutorials: Many users face difficulties at initial phase of operating or utilizing the AR applications. AR designers suggested that implementing step by step guidance may help user to learn at their own pace. In addition, interactive tutorials about how to use AR applications can definitely help to better understand the functionality of complex AR systems.

We need to rethink how we introduce users to AR apps. The onboarding process should be fun and easy to understand. (S2)

Enhancing visual clarity and minimizing overlap: Most of the AR applications fails in offering proper visual aspects that has ruin the overall experience. UX designer suggested that color contrast, shadows, subtle animations are the minimal visual elements that help virtual objects stand out from the real-world background. This would make it easier for users to interact with AR elements, especially in environments with complex or noisy backgrounds.

"We need to be mindful of how AR elements blend with real-world environments. Using clear visual separation techniques would help users navigate more confidently." (S4)

User-centered customization options: Several professionals point out the importance of user-centered customization options, such as adjusting UI layouts, font sizes, color theme or interaction modes. The one common voice was to offer users with option for hiding and showing certain elements and switching gesture control can significantly enhance the accessibility and usability of AR apps.

"Every user is different. If we allow them to have full control over their experience than only, we can say we built a user-friendly application." (S3)

Improving gesture recognition and interaction method: Participants suggested that alternative input methods such as voice commands, physical controllers should be provided in AR applications. This can be helpful whenever other gestures are not recognized correctly.

"As it is said that we must have a backup plan: if option A fails, then there is always option B. This way, users are not frustrated by a single point of failure and can seamlessly switch to alternative methods to achieve their goals." (S1)

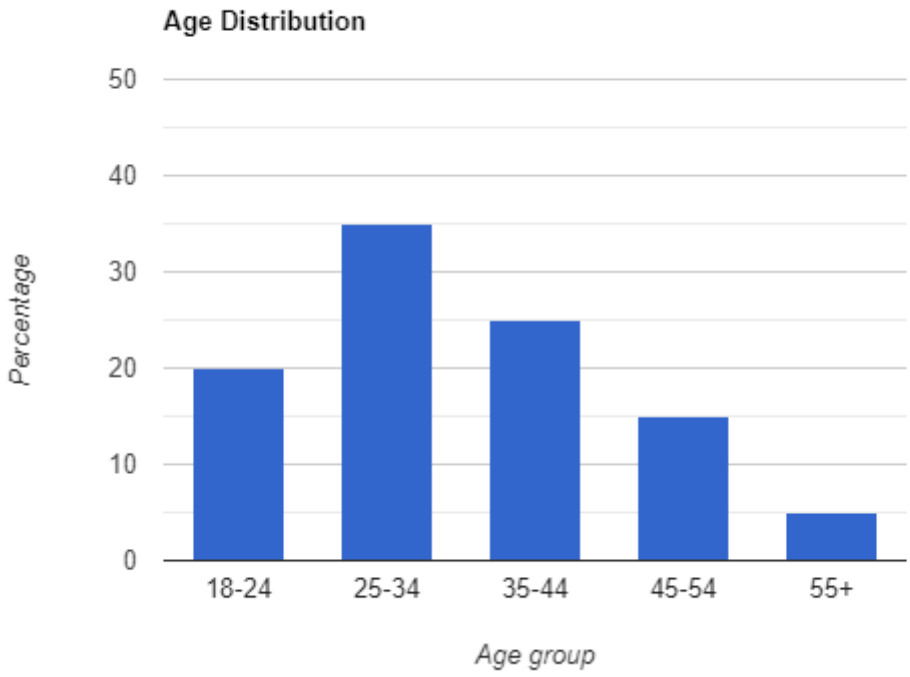
4.2 Quantitative findings from User Surveys

This section outlines the quantitative findings from our survey on enhancing user experience in AR applications focusing on interface design. As mentioned previously, the survey was conducted among approximately 50 AR applications user. The results are categorized based on several key areas, such as demographic info, AR usage patterns, interface preferences, feature importance and overall satisfaction.

4.2.1 Demographic info

It is essential to know the demographic profiles of AR users before designing the interface in order to understand their needs and design effectively. Our survey has captured user's age distribution to provide valuable insights into the user base of AR applications. This results reveal that the majority of AR applications users are between the ages of 25-44 years. The age range from 25-34 years holds the highest segment (35%) of respondents. This findings suggests that AR applications are well popular and mostly used by young to middle aged adults. The lower representation in the 55+ category (5%) indicates potential opportunities for making AR interfaces more accessible and appealing to older users. The result is shown as below.

Figure 1: Age distribution of survey respondents



4.2.2 Most frequently used AR applications and their usage patterns

To design effective AR interface, it is essential to understand how frequently users interact with AR applications and which applications they use the most. The survey results shows that the large number (40%) of respondents use AR applications at least weekly. Similarly, it is followed by 30% of respondents who use AR applications in daily basis. This high frequency of interaction tells the importance of designing intuitive and efficient interfaces that can support regular use without causing any problems or frustration. The following figure presents the distribution of AR application usage frequency:

Figure 2: AR application usage frequency

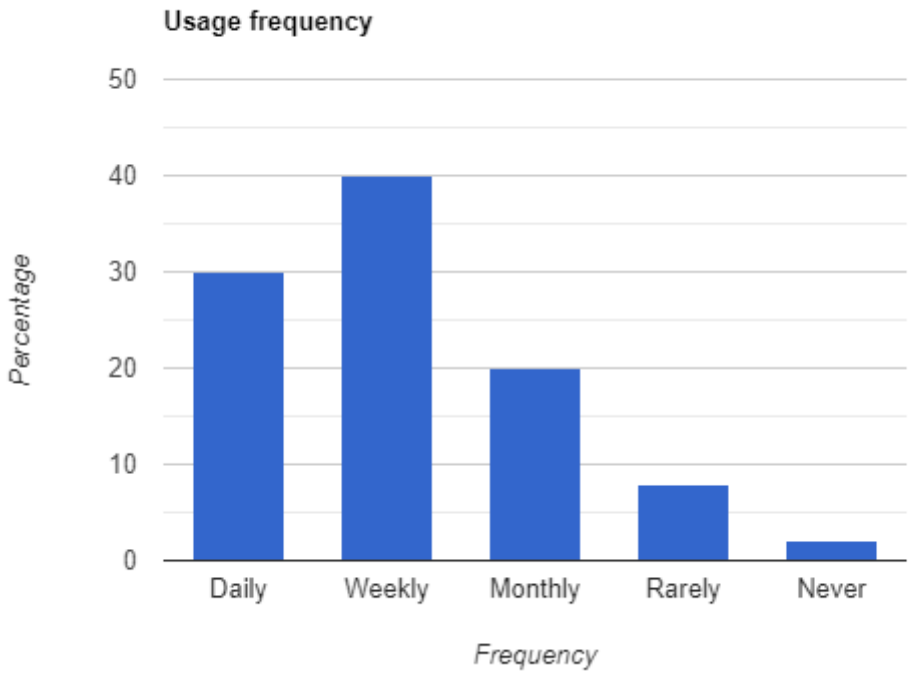
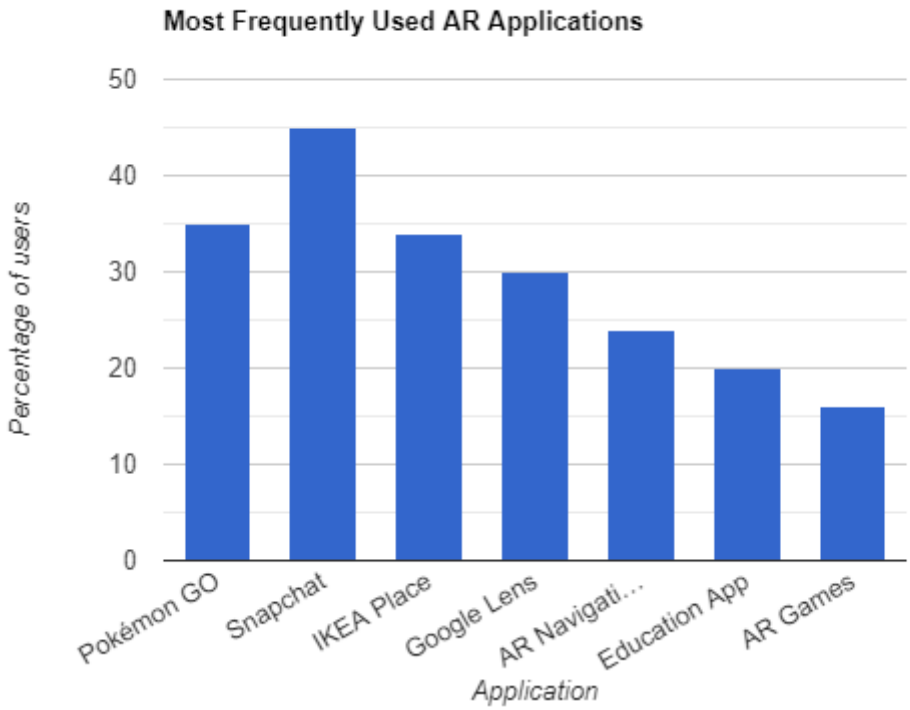


Table 1: Age distribution and Usage frequency of AR application

Age Group	Dailly Use	Weekly Use	Monthly Use	Rarely Use	Never Use
18-24 years	20%	25%	30%	15%	10%
25-34 years	30%	35%	20%	10%	5%
35-44 years	10%	20%	35%	25%	10%
45-54 years	5%	15%	30%	35%	15%
55+ years	2%	5%	15%	40%	38%

According to our respondents, Pokémon Go and Snapchat filters are still quite popular nowadays. However, there is also significant adoption of utility apps like Ikea place and google lens, which are also being used in day today life.

Figure 3: Most frequently used AR applications



4.2.3 AR Interface Elements

To design an interface, the first and foremost important thing is to have clear understanding about the needs and preferences of users. In AR systems, different elements and methods are used for input commands, processing them, and generating outputs accordingly. There are several control elements which are already been implemented and some of them are still being upgraded. So, with the aim of understanding the satisfaction level of current interaction methods, we have conducted user survey, and we got the following results.

Table 2: User preferences for AR interface elements

Element	Count	Percentage
Gesture controls	35	70%
On-screen button	33	66%
Eye tracking	15	30%
Voice commands	12	24%

Gesture controls and on-screen buttons are the most intuitive elements preferred by 70% and 66% of respondents respectively. This results says that the current UI design principles are effectively incorporating these familiar interactions methods. However, eye tracking and voice command are the key features and strong approach to redefine the user interaction in AR applications.

4.2.4 Feature Importance

Currently, there are many AR applications available, and each with different features, layout and purposes. However, there are four key features which are extremely important to consider when designing AR application: responsiveness, visual clarity, ease of navigation, and customization. Identifying which features users value most can definitely help us to prioritize and design AR interface. From our survey results, it is clear that user prefer to have adoptive layouts with minimal UI design. This has shed light on the need for AR interfaces to be highly reactive and visually precise, ensuring a seamless integration with the real world.

Table 3: Average importance ratings of AR features

Feature	Average Importance (1-5)
Responsiveness	4.4
Visual clarity	4.2
Ease of navigation	3.8
Customization	3.5

4.2.5 User Challenges

We have asked our respondents, what is the biggest challenge you face when using AR applications? And the responses that we received likely highlight areas that should draw the attention of UX designers and AR experts when designing user interfaces. The common challenges reported by AR users includes:

- Physical discomfort during prolonged use: After using AR applications for longer period of time has caused several physical discomfort. Such as, eyes strain, motion sickness, physical fatigue.
  "The Ikea place app is great, but focusing on AR furniture for long time has cause a headache." (R2, age 41)
  "I love the concept of AR navigation apps, but the constant movement of AR directions makes me dizzy, especially in a moving car." (R12, age 35)
- Technical limitations: Majority of respondents face technical issues and said that they typically encounter common problems. For example, battery drain, phone heating up, lags on older phone, some apps are not compatible with android devices.
  "Using AR features in social media apps like Instagram and Snapchat significantly reduces my phone's battery life." (R42, age 27)
  "I tried to use an AR interior design app, but it wasn't compatible with my phone's camera." (R24, age 36)
- Learning curve and accessibility: Another issue reported by respondents is that users with limited knowledge about AR often get lost and could not figure out what to do during onboarding process or how to properly use the apps. Similarly, users have stated that some AR game apps rely on colour-based interactions, which poses a big problem for people with partial colour blindness.
  "The text in many AR apps is too small on my phone screen. It's particularly challenging for AR information overlays." (R33, age 48)
- Unintuitive controls and information overloaded: One respondent stated that the input controls vary from one applications to another, and it is very difficult to understand.

“In some AR games, it's not clear how to interact with virtual objects. I often find myself tapping randomly on the screen.” (R14, age 28)

5 Discussion

In this section, we will interpret the results that we obtained from findings chapter with our research questions.

5.1 Current user interface design principles in AR applications

The landscape of AR user interface design is rapidly evolving with several key principles emerging as crucial for creating effective and engaging AR experiences. Several previous studies have discussed various design principles for the user interface of augmented reality. For example, Bloksa (Blokša, 2017), Shukri (Shukri et al., 2017) and Oluwadunsin Dabor (Dabor O, 2019). has come up with a proposal of user interface design for AR user interface. Likewise, recently Apple has also shared some design guidelines(Martina Sartor, 2023) which are more specific with Apple's vision pro. However, our research has uncovered four primary design principles which are currently applied in AR user interfaces.

The principle of minimalism and simplicity stands out as a cornerstone of effective AR interface design. Both designers and users in our study emphasized the critical importance of avoiding cluttered interfaces and displaying only essential content. This approach is particularly crucial in AR environments where users must simultaneously interact with both virtual and physical elements. The preference for "adoptive layouts with minimal UI design" expressed by users in our survey underscores the significance of this principle. By embracing minimalism, AR designers can create interfaces that complement rather than compete with the real-world environment, thereby enhancing the overall user experience and reducing cognitive load.

Spatial awareness emerged as another key principle, with designers highlighting the importance of creating AR elements that feel integrated with the physical world. This principle aims to enhance the user's sense of connection between virtual and physical spaces, potentially reducing cognitive load and improving overall user experience. By carefully considering the spatial relationships between digital content and the real world, designers can create more intuitive and immersive AR experiences. This principle aligns with the broader goal of making AR

interactions feel natural and seamless, as if the virtual elements are a genuine extension of the user's physical environment.

The principles of real-time feedback and visual clarity round out the core design tenets identified in our research. Quick response or notifications is one of the critical aspects of AR UI design, which is also highlighted as important feature in our survey. This helps us to create a more engaging and interactive AR experience, allowing users to feel a direct connection between their actions and the digital content. Visual clarity, rated as the second most important feature by users, further supports this goal. Designers suggested techniques such as using colour contrast, shadows, and subtle animations to help virtual objects stand out from real-world backgrounds, ensuring that AR elements are easily distinguishable and interactable within the mixed reality environment.

5.2 Challenges face by users with existing AR UIs

Despite the progress made in AR interface design, users continue to face several significant challenges when interacting with AR applications. Our study identified six key areas of difficulty: interface usability, gesture recognition issues, physical discomfort, technical limitations, cognitive load and multitasking, and learning curve and accessibility. These issues create complexity while crafting seamless and smooth user experiences in AR and highlights the necessity of continued innovation in AR UI design.

Several designers mentioned that AR interfaces tended to get complex and confusing, which affected the ease of use and task accomplishment. That said, users' testimonies highlighted issues such as information overload and illogical controls. The major issue in designing an application is to decide on what aspects need to be included and what should be skipped to avoid cluttering the interface. This is especially important in AR where the interface is placed within the real world and must not overpower or interfere with it.

Gesture recognition issues presented another significant hurdle for AR users. Both designers and users reported difficulties with gesture-based interactions, citing inconsistency and

unresponsiveness as major issues. The difficulty arises due to the widespread promotion of gesture controls as a simple and natural way to engage with AR experiences. However, the effectiveness of gesture recognition can be varied based on different tracking method used. For example, hand-held tracking devices are the dedicated with sensors which provide better precision because they can detect human's hand movements accurately. This type of devices is mostly purposed to built AR experiences but at the same time, it might limit the user's freedom of movement and constantly holding a device for long time could be inconvenient. On other hand, camera-based gesture recognition used phone or headset camera for tracking the hand movement through image recognition. However, this could be more manageable as it enables hands free interaction. But due to different external factors such as lighting, background, and positioning can affect its accuracy and possibly misread or does not detect at all. This inconsistency can result in user frustration, as users expects seamless interactions. Moreover, users indicated they encountered physical distress including eyes strain motion sickness and fatigue when engaging with AR applications over time. The physical challenges reduce the user's enjoyment and could restrict the uptake of AR technology significantly for apps that demand lengthy engagement.

Technical limitations, cognitive load, and accessibility issues are the major challenges identified in our study. Users frequently reported problems such as battery drain, device heating, and compatibility issues, highlighting the hardware constraints that still impact AR experiences. The cognitive demands of AR interfaces also proved challenging, with users struggling to process both virtual and physical information simultaneously. This multitasking requirement can lead to mental fatigue and reduced effectiveness, particularly in complex AR applications. Lastly, the learning curve associated with AR applications poses a significant barrier, especially for users with limited AR knowledge. Accessibility issues were also noted, particularly for users with visual impairments, underscoring the need for more inclusive AR design practices. Visual impairments brought attention to accessibility challenges which reinforce the necessity of designing all-inclusive AR applications. These complications signify the early development of AR technology and emphasize the major tasks to form intuitive and accessible interfaces.

5.3 Design approaches or solutions

As a consequence of the findings from our investigation several innovative design solutions and techniques have surfaced. These methods intend to resolve the fundamental challenges encountered by AR users and designers while boosting usability and user experience. The proposed methods cover a mix of techniques from refined onboarding practices to user-friendly interfaces.

Enhanced onboarding and tutorials stand out as a crucial strategy for improving the AR user experience. By implementing step-by-step guidance and interactive tutorials, designers can help users better understand AR functionality and navigate the unique challenges of mixed reality environments. This method tackles the practical learning challenges we found in our study. With a goal of boosting visual clarity through increased colour contrast and animations an enhanced visual design appeared as a crucial method. These techniques can help distinguish AR elements from the real-world background, addressing the challenges of information overload and interface clutter reported by users.

User-centered customization offers another promising avenue for improving AR interfaces. Users can experience increased accessibility and usability through the provision of options to change UI layouts font sizes and colour themes as well as interaction modes. This method understands the various requirements and choices of AR users and enables a personalized and possibly more enjoyable AR experience. Creating alternative input solutions like voice instructions and physical tools can further the use of gestures while resolving the reliability and responsiveness concerns noted with today's gesture recognition tools.

The concept of adaptive interfaces represents a more advanced solution to the challenges of AR UI design. By developing interfaces that can adjust based on the user's environment and level of expertise, designers can address the challenge of user experience variability and potentially reduce cognitive load. This approach could involve dynamically simplifying interfaces for novice users or in complex environments or offering more advanced features as users become more proficient. Lastly, a renewed focus on ergonomics in AR design could help mitigate the physical discomfort reported by users. This might involve incorporating features that encourage breaks

during extended use or adjusting visual elements to reduce eye strain. By considering the physical impact of AR use, designers can create more comfortable and sustainable AR experiences, potentially encouraging longer-term adoption of the technology.

6 Conclusions

In conclusion, this research examines the current state of augmented reality user interface design by emphasizing key design concepts and proposing solutions to user difficulties. This study used both qualitative interviews with designers and quantitative surveys of users to acquire vital insights into the present state of AR UI design and user experiences. This research has contributed to the knowledge base for AR user interface design by identifying important concepts, highlighting user difficulties, and proposing potential solutions. The outcomes of this study can provide designers and developers with vital information for creating user-friendly and engaging AR applications. The possibilities for AR are tremendous and by tackling the issues raised in this study, we may explore the full potential of this new technology.

6.1 Research Contribution

The primary contributions of this research are threefold:

First, this study has identified and elaborated on four key design principles currently applied in AR user interfaces:

- minimalism and simplicity
- spatial awareness
- real-time feedback
- visual clarity

By integrating both designer expertise and user needs into their strategies experts have built guidelines for effective and user-friendly AR interfaces. Concentrating on simplicity and spatial understanding distinguishes the distinctive difficulties of creating for combined reality settings where digital and real-world components need to merge smoothly.

Second, this research has revealed the key difficulties for users utilizing AR applications. This study has revealed problems like,

- interface usability issues and
- gesture recognition inconsistencies

that contradict user requirements in current AR interfaces. These results offer important guidance to both researchers and designers in the area of augmented reality.

Third, this study provided a number of design methods and solutions to address the highlighted difficulties. These recommendations, which range from improved onboarding processes to adaptive interfaces and practical considerations, provide actionable strategies for improving AR user experiences. By relating these solutions to user-reported issues, this study lays the foundation for the next generation of AR interface design.

6.2 Limitation

While this study offers useful insights on AR UI design, it is essential to understand its limits. The sample size, particularly for the quantitative survey (about 50 respondents), was limited, perhaps limiting the generalizability of the findings. Furthermore, the study's demographic profile shows a preference for users aged 25-44, with little representation of senior users. This demographic point may not accurately reflect the experiences and preferences of a larger user group.

Furthermore, the study focuses on currently accessible AR applications, which may not completely represent developing trends or experimental interfaces that are still being developed. Due to the quick speed of technological innovation in the AR area, several of the listed difficulties may already be overcome in newer applications or technology.

6.3 Future Work

This study's findings and limitations suggest many interesting pathways for future research:

Cross-platform Comparisons: It would be useful to investigate how AR UI design concepts and challenges differ across different devices (for example, smartphones, headsets, and smart

glasses). This approach may result in multiple design standards that serve for the unique capabilities and challenges of various AR mediums.

Accessibility in AR: Additional research into creating AR interfaces for users of all abilities needs to be done with a focus on the issues that users with visual impairments encounter. This research could lead to more inclusive AR design methods and a larger potential user base for AR apps.

Integration of Emerging Technologies: It's interesting to investigate how technology such as physical input and eye tracking might improve AR interaction design. These technologies have the potential to solve some of the current limitations in AR interaction, such as inconsistent gesture recognition.

Cognitive Load Optimization: Further research on techniques for managing cognitive load in AR environments is important. It may be necessary to develop alternative ways showing information and to create adaptable interfaces that match users' cognitive skills.

7 References

Alkhamisi, A. O., & Monowar, M. M. (2013). Rise of Augmented Reality: Current and Future Application Areas. *International Journal of Internet and Distributed Systems*, *01*(04), 25–34. https://doi.org/10.4236/ijids.2013.14005

Arena, F., Collotta, M., Pau, G., & Termine, F. (2022). An Overview of Augmented Reality. In *Computers* (Vol. 11, Issue 2). MDPI. https://doi.org/10.3390/computers11020028

Billinghurst, M., Clark, A., & Lee, G. (2015). *A Survey of Augmented Reality*. now Publishers Inc. https://doi.org/10.1561/9781601989215

Blokša, J. (2017). *Design Guidelines for User Interface for Augmented Reality. Masaryk University*

Bracho Mosquera, A. S. (2024). Applications of augmented reality in museums, impact on cultural heritage. *Gamification and Augmented Reality*, *2*, 34. https://doi.org/10.56294/gr202434

Challenor, J., & Ma, M. (2019). A Review of Augmented Reality Applications for History Education and Heritage Visualisation. *Multimodal Technologies and Interaction*, *3*(2), 39. https://doi.org/10.3390/mti3020039

Cruz-Carrizales, C.-A., Bautista-Rojas, L.-E., & Pedraza-Ferreira, G.-R. (2021). Evaluation of User Interfaces Description Languages to Augmented Reality. *Revista Facultad de Ingeniería*, *30*(57), e13304. https://doi.org/10.19053/01211129.v30.n57.2021.13304

Davidavičienė, V., Raudeliūnienė, J., & Viršilaitė, R. (2020). EVALUATION OF USER EXPERIENCE IN AUGMENTED REALITY MOBILE APPLICATIONS. *Journal of Business Economics and Management*, *22*(2), 467–481. https://doi.org/10.3846/jbem.2020.13999

Ejaz, A., Asim Ali, S., Yasir Ejaz, M., & Ahmed Siddiqui, F. (2019). Graphic User Interface Design Principles for Designing Augmented Reality Applications. In *IJACSA) International Journal of Advanced Computer Science and Applications* (Vol. 10, Issue 2). www.ijacsa.thesai.org

Galletta, A. (2013). *Mastering the Semi-Structured Interview and Beyond. New York University Press*

Grubert, J., Langlotz, T., Zollmann, S., & Regenbrecht, H. (2017). Towards Pervasive Augmented Reality: Context-Awareness in Augmented Reality. *IEEE Transactions on Visualization and Computer Graphics*, *23*(6), 1706–1724. https://doi.org/10.1109/TVCG.2016.2543720

Jung, T. H., & tom Dieck, M. C. (2017). Augmented reality, virtual reality and 3D printing for the co-creation of value for the visitor experience at cultural heritage places. *Journal of Place Management and Development*, *10*(2), 140–151. https://doi.org/10.1108/JPMD-07-2016-0045

Li, W., & Cheng, S. (2023). Research on the influence of using augmented reality technology in industrial display design. In L. Xiao & J. Xue (Eds.), *Fourteenth International Conference on Graphics and Image Processing (ICGIP 2022)* (p. 129). SPIE. https://doi.org/10.1117/12.2680394

Lim, K. Y. T., & Lim, R. (2020). Semiotics, memory and augmented reality: History education with learner-generated augmentation. *British Journal of Educational Technology*, *51*(3), 673–691. https://doi.org/10.1111/bjet.12904

Malta, A., Farinha, T., & Mendes, M. (2023). Augmented Reality in Maintenance—History and Perspectives. *Journal of Imaging*, *9*(7), 142. https://doi.org/10.3390/jimaging9070142

Md Nor, R., & Abdul Razak, M. H. (2021). Interactive Design in Enhancing User Experience in Museum. *Journal of Computing Research and Innovation*, *6*(3), 86–91. https://doi.org/10.24191/jcrinn.v6i3.249

Martina Sartor. (2023). *Mastering Apple's AR Guidelines: An Exploration into Designing for AR*.

Nielsen J. (1993). *Usability Engineering*. Elsevier. https://doi.org/10.1016/C2009-0-21512-1

Nigam, S., & C, Prof. S. (2022). Augmented Reality in Education System. *International Journal for Research in Applied Science and Engineering Technology*, *10*(7), 397–401. https://doi.org/10.22214/ijraset.2022.45202

Öztürk, B., & Genç, Y. (2023). Using Augmented Reality Interfaces for Artificial Intelligence Systems. *Soft Computing, Artificial Intelligence and Applications*, 207–222. https://doi.org/10.5121/csit.2023.132416

Pantano, E., Rese, A., & Baier, D. (2017). Enhancing the online decision-making process by using augmented reality: A two country comparison of youth markets. *Journal of Retailing and Consumer Services*, *38*, 81–95. https://doi.org/10.1016/j.jretconser.2017.05.011

Putro, H. T., & Setyowati, E. (2022). Development of Application Based on Augmented Reality as A Learning of History and Culture in Architecture Case Study Pathok Negoro Mosques Yogyakarta. *Journal of Artificial Intelligence in Architecture*, *1*(1), 1–9. https://doi.org/10.24002/jarina.v1i1.4835

R. Silva, J. C. O. G. A. G. (n.d.). *Introduction to Augmented Reality*.

Robson, C. (2011). *Real world research: A resource for users of social research methods in applied settings* (3rd ed.).

Tenh, H. K., & Shiratuddin, N. (2022). Components of Adaptive Augmented Reality Model for Heritage Mobile Application. *International Journal of Interactive Mobile Technologies (IJIM)*, *16*(02), 17–27. https://doi.org/10.3991/ijim.v16i02.27317

Wen, Y., Wu, L., He, S., Ng, N. H.-E., Teo, B. C., Looi, C. K., & Cai, Y. (2023). Integrating augmented reality into inquiry-based learning approach in primary science classrooms. *Educational Technology Research and Development*, *71*(4), 1631–1651. https://doi.org/10.1007/s11423-023-10235-y

Williams, M., Yao, K. K. K., & Nurse, J. R. C. (2020). *Developing an Augmented Reality Tourism App through User-Centred Design (Extended Version)*.

8 Appendices

Appendix 1: Interview Questions

Section 1: Background and Expertise

1. Can you briefly describe your background and experience in AR design?
2. What types of AR applications have you worked on (e.g., gaming, retail, navigation, education)?

Section 2: Current Design Approaches in AR UI

3. How would you describe your design approach or methodology for creating user interfaces in AR?
4. What specific design principles do you prioritize when creating AR interfaces? (e.g., usability, accessibility, intuitiveness)
5. How does your approach differ when designing for AR compared to traditional 2D or mobile interfaces?

Section 3: Key Design Challenges in AR

6. What are the biggest challenges you encounter when designing user interfaces for AR applications?

Section 4: User-Centric Design and Feedback Loops

7. How often do you gather user feedback during the design process, and at what stages?

Section 5: Future Perspectives

8. Are there any tools or resources you find particularly helpful for designing AR experiences?
9. What advice would you give to designers entering the AR space on how to approach UI challenges effectively?

Appendix 2: Survey Form

1. What is your age group?

a) 18-24 b) 25-34 c) 35-44 d) 45-54 e) 55+

2. How often do you use AR applications?

a) Daily b) Weekly c) Monthly d) Rarely e) Never

3. Which device(s) do you typically use for AR? (Select all that apply)

a) Smartphone b) Tablet c) AR Glasses d) Others

4. Which AR applications do you use most frequently? (Select all that apply)

a) Pokémon Go b) Snapchat c) IKEA Place d) Google Lens e) AR Navigation apps
f) Education apps g) AR Games h) Others

5. On a scale of 1-5, how comfortable are you with using AR interfaces?
6. How easy is it to understand the visual cues or instructions in the AR application?
   a) Very easy b) Easy c) Neutral d) Difficult e) Very Difficult
7. What types of interaction methods have you used in AR? (Select all that apply)
   a) Touchscreen b) Gestures c) Voice Commands d) Eye-tracking
8. What challenges have you faced in using AR applications? (Select all that apply)
   a) Confusing navigation or controls d) Fatigue or discomfort over extended use
   b) Difficulty interacting with virtual objects e) Lack of clear instructions or feedback
   c) Poor visual clarity or field of view
9. What improvements would you like to see in AR interfaces?
   a) More intuitive controls b) Clearer instructions or feedback c) Better Visual quality d) Improved comfort and ergonomics e) Lack of clear instructions or feedback f) Enhanced interaction options (e.g., better gesture or voice recognition)
10. How likely are you to recommend AR applications to others?
11. Any additional comments on how AR applications could improve their user interface and experience?